

THE MIDNIGHT MISSION

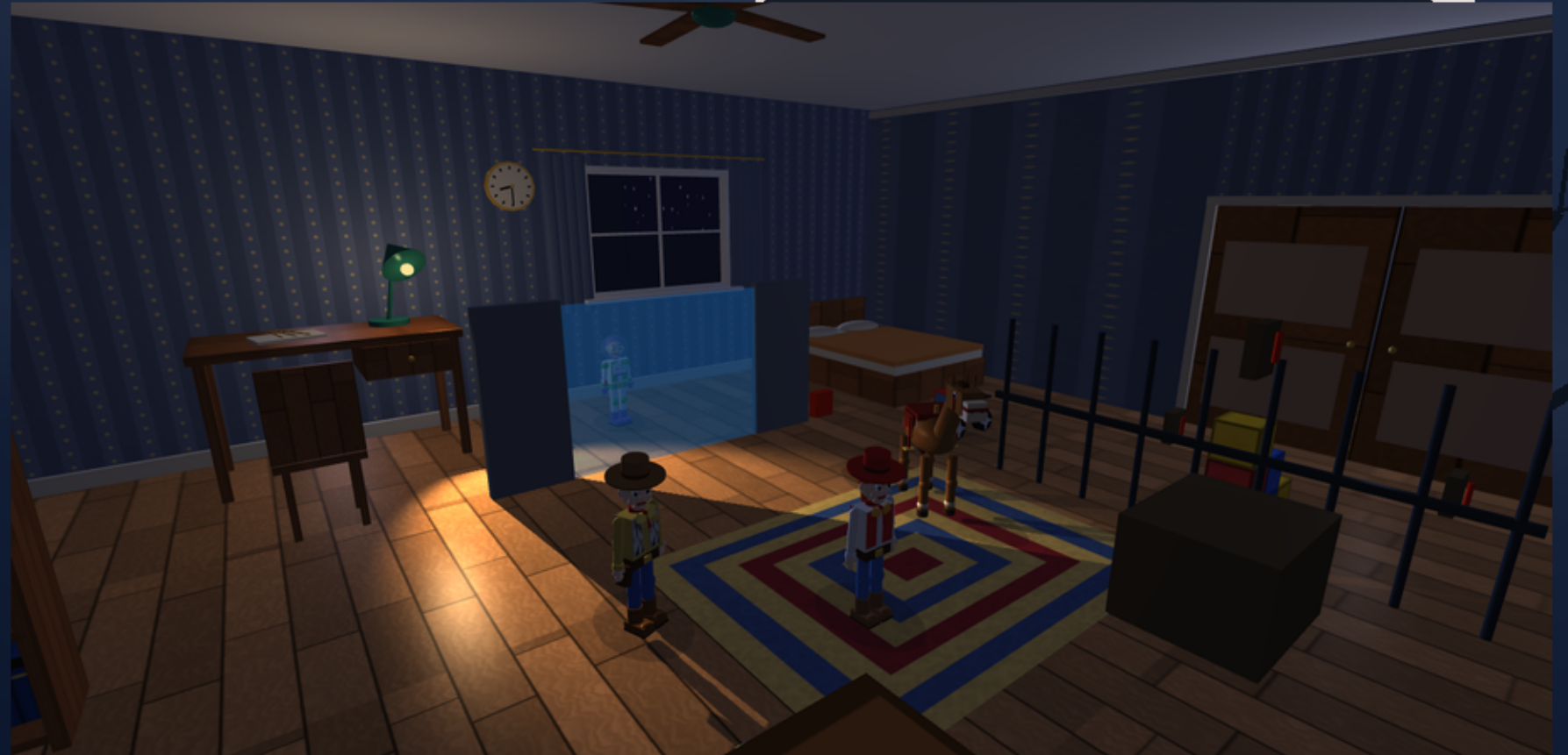
A haunted toy room. An interactive graphics story.

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CSE-4102 · Computer Graphics and Image
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ROOM NIGHT

EXPLORE, CONTROL, COMPARE

The proposal is realised as a complete arrival, puzzle, rescues and outdoor escape.

Five driveable toys; ball and lamp controls

TRS + hierarchy; mounting and dismounting

Three light types; Gouraud and Phong

22 surface maps; analytic BVH ray tracing

PLAY PROJECT DEMO



HOUSE



MOUNTED

PRIMITIVES BECOME CHARACTERS

Position, normal and UV data define five indexed mesh families.

Plane 4 vertices / 2 triangles; cube 24 / 12

Sphere: $24 \times 36 \rightarrow 925$ vertices / 1656 triangles

Cylinder: 134 / 128; cone: 99 / 64

The scene uses shared buffers and instance transforms



WOODY



BUZZ



PENNY

A WORLD WITH SURFACE DETAIL

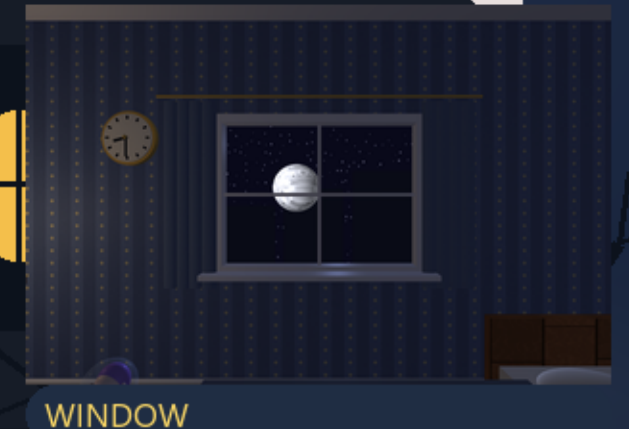
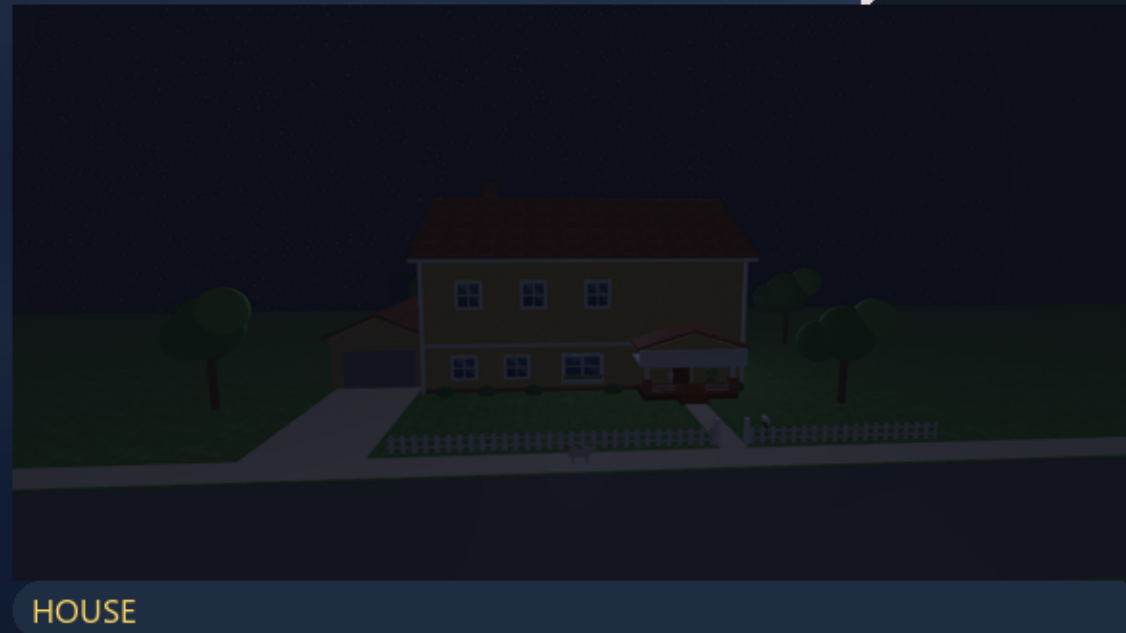
The house, garden, room and furniture belong to one connected scene.

Wood grain, wallpaper, cloth, moon craters and book rows

UV repeats multiply material colour by sampled RGB

Picket alpha cuts holes and replaces repeated geometry

22 maps plus white in one texture array serves the ray tracer



ONE PARENT MOVES THE RIDER

$$M_{\text{world}}(\text{child}) = M_{\text{world}}(\text{parent}) \times M_{\text{local}}(\text{child})$$

R: mount/dismount within 2.2 units

1-5 + W/S/A/D: choose and drive a toy

Select: live takeover; 0 release; N full manual

F: inspect; C: Free / Orbit / Follow

Normal matrix = inverse transpose of world linear part



LIVE CONTROL



MOUNTED

EVERY MOTION HAS A RULE

Delta time connects displacement, gait, wheel rotation and camera motion.

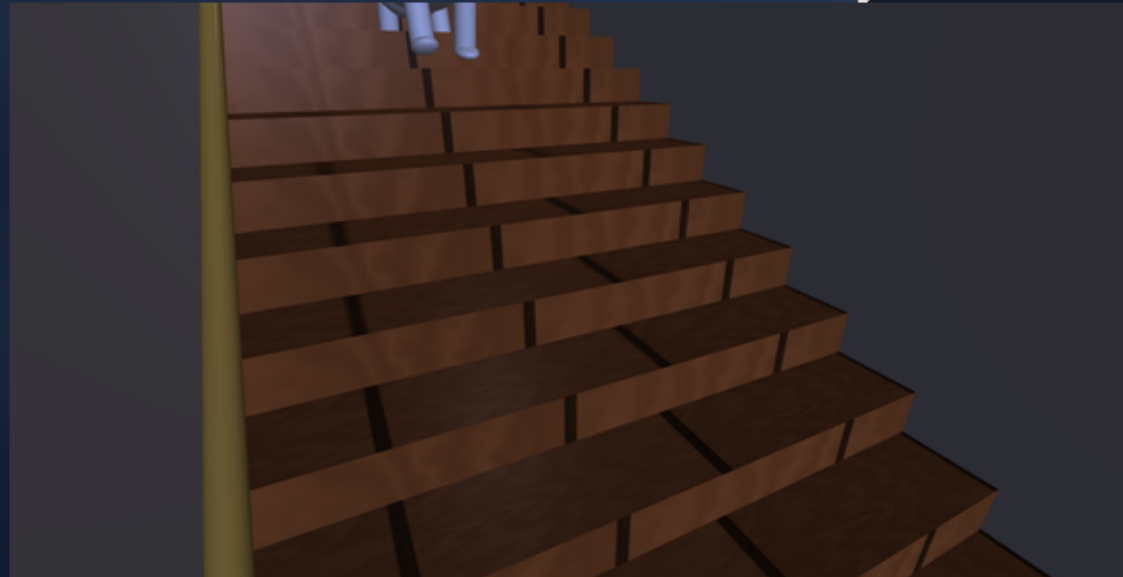
Gait angle = $\sin(\text{distance phase}) \times \text{amplitude}$

Wheel angle = $\text{distance} / \text{radius}$

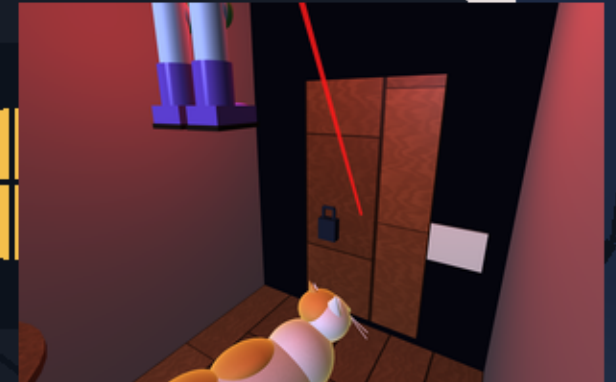
Doors rotate at hinges; Penny follows 3D waypoints

O: ghost, ball and lamp ambience

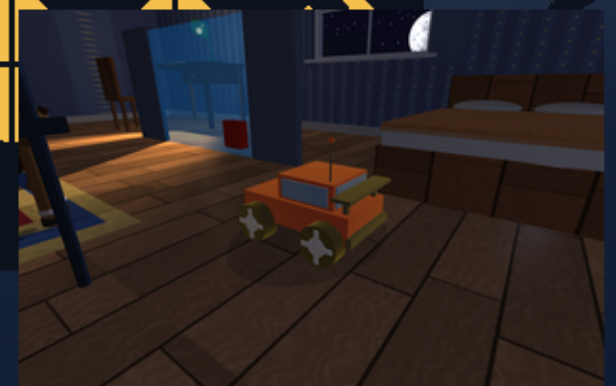
Fan rotor: $110^\circ/\text{s}$; clock hands follow scene hour



STAIR DESCENT



DOOR IMPACT



CAR

THREE LIGHT TYPES

Ambient + diffuse + specular + emission

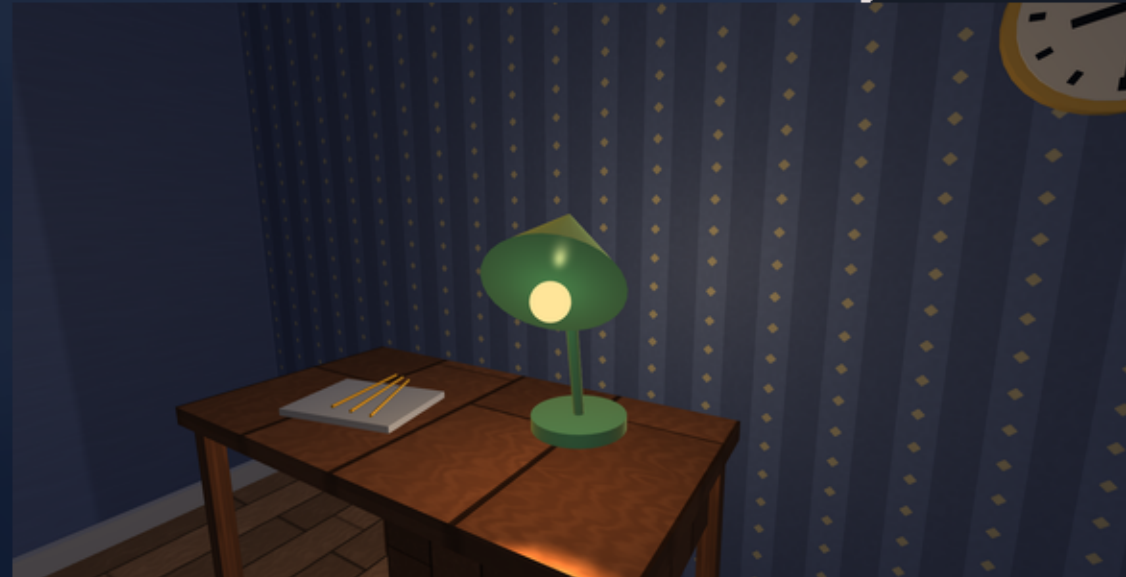
Directional: sun/moon; parallel rays

Point: lamp bulb; $1/(k_c + k_l d + k_q d^2)$

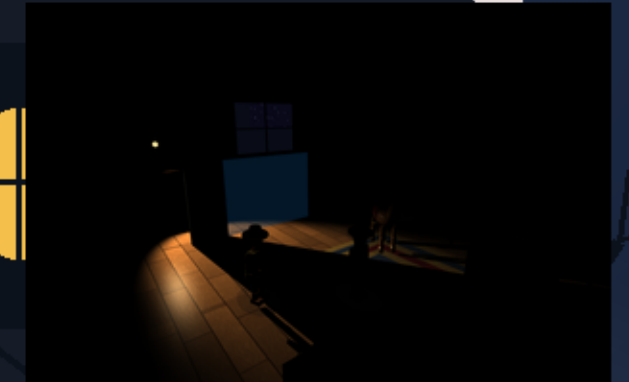
Spot: lamp/headlights; smooth inner/outer cone

Diffuse = $k_d \max(N \cdot L, 0)$

Specular = $k_s \max(R \cdot V, 0)^n$



LAMP



SPOT



DIFFUSE

WHERE IS LIGHT EVALUATED?

Matched views expose the difference between vertex and pixel illumination.

Flat: triangle normal from position derivatives

Gouraud: vertex lighting → interpolated colour

Phong: interpolated normal → pixel lighting

Blinn: half-vector $H = \text{normalize}(L+V)$

F2 cycles raster modes; F4 returns to ray tracing



FLAT



GOURAUD



PHONG



BLINN

RAYS REVEAL A SECOND VIEW

Primary ray → nearest analytic hit → illumination → bounded continuation

BVH median split; world-space boxes;
near child first

Five analytic primitive intersection
equations

Shadow rays test visibility to selected
lights

Mirror floor/ball; straight transparency

Raster lamp: 2048^2 depth map + 3×3
PCF



RAY ZERO



ROOM RAY

10 THE TOYS ARE SAFE

THANK YOU

All five escape. The scene remains open to inspection.

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CSE-4102

Geometry · hierarchy · motion ·
interaction

Lighting · shading · textures · ray tracing



STORY END

